

Database Design Principles Personal Project

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-For this Project I selected a scenario, Identified it's entities and attributes and Wrote SQL and BI queries.

Scenario Selection

-Gym Membership and Class Tracking

- Members
- Membership plans
- Fitness classes
- Instructors
- Attendance tracking
- Payments

-I then Listed the Entities and Attributes

Entities & Attributes

-Entities & Attributes

- **Member** — MemberID, FirstName, LastName, Email, Phone, JoinDate, MembershipPlanID
- **MembershipPlan** — PlanID, PlanName, MonthlyFee, Description
- **Instructor** — InstructorID, FirstName, LastName, Specialty
- **FitnessClass** — ClassID, ClassName, InstructorID, ScheduleDay, ScheduleTime
- **ClassAttendance** — AttendanceID, MemberID, ClassID, AttendanceDate
- **Payment** — PaymentID, MemberID, PaymentDate, Amount

-I then wrote Secured Query Language(SQL) Code that will create the tables needed Also BI Queries

SQL Code that creates tables

```
CREATE TABLE MembershipPlan (  
    PlanID AUTOINCREMENT PRIMARY KEY,  
    PlanName TEXT(50),  
    MonthlyFee CURRENCY,  
    Description TEXT(255)  
);
```

```
CREATE TABLE Member (  
    MemberID AUTOINCREMENT PRIMARY KEY,  
    FirstName TEXT(50),  
    LastName TEXT(50),  
    Email TEXT(100),  
    Phone TEXT(20),  
    JoinDate DATE,  
    MembershipPlanID LONG,  
    FOREIGN KEY (MembershipPlanID) REFERENCES MembershipPlan(PlanID)  
);
```

```
CREATE TABLE Instructor (  
    InstructorID AUTOINCREMENT PRIMARY KEY,  
    FirstName TEXT(50),  
    LastName TEXT(50),  
    Specialty TEXT(50)  
);
```

```
CREATE TABLE FitnessClass (  
    ClassID AUTOINCREMENT PRIMARY KEY,
```

```
ClassName TEXT(50),
InstructorID LONG,
ScheduleDay TEXT(20),
ScheduleTime TEXT(20),
FOREIGN KEY (InstructorID) REFERENCES Instructor(InstructorID)
);
```

```
CREATE TABLE ClassAttendance (
AttendanceID AUTOINCREMENT PRIMARY KEY,
MemberID LONG,
ClassID LONG,
AttendanceDate DATE,
FOREIGN KEY (MemberID) REFERENCES Member(MemberID),
FOREIGN KEY (ClassID) REFERENCES FitnessClass(ClassID)
);
```

```
CREATE TABLE Payment (
PaymentID AUTOINCREMENT PRIMARY KEY,
MemberID LONG,
PaymentDate DATE,
Amount CURRENCY,
FOREIGN KEY (MemberID) REFERENCES Member(MemberID)
);
```

BI Queries (Access SQL)

-Total Revenue by Month

```
SELECT
    FORMAT(PaymentDate, "yyyy-mm") AS RevenueMonth,
    SUM(Amount) AS TotalRevenue
FROM Payment
GROUP BY FORMAT(PaymentDate, "yyyy-mm")
ORDER BY RevenueMonth;
```

-Most popular by classes

```
SELECT
    FitnessClass.ClassName,
    COUNT(ClassAttendance.AttendanceID) AS AttendanceCount
FROM FitnessClass
LEFT JOIN ClassAttendance
    ON FitnessClass.ClassID = ClassAttendance.ClassID
GROUP BY FitnessClass.ClassName
ORDER BY AttendanceCount DESC;
```

-Members by membership plan

```
SELECT
    MembershipPlan.PlanName,
    COUNT(Member.MemberID) AS MemberCount
FROM MembershipPlan
LEFT JOIN Member
    ON MembershipPlan.PlanID = Member.MembershipPlanID
GROUP BY Membersh
```

Reflection

For this project, my goal was to apply database design principles. This reflection explains what I did and what I learned from the experience. I worked through the project step by step I started with simple thoughts like what the entities and attributes are. I wrote BI queries that would help analyze revenue, attendance, and membership trends. This is clear, organized information that makes it easier to move from planning to implementation. I identified the right entities for a gym system and proposed relationships that make sense. I have created a good database design using database principles and practices.